

Runge–Kutta Random Feature Method for Solving Multiphase Flow Problems of Cells

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1 Introduction

Cell collective migration plays a crucial role in many physiological processes, with growing interest in understanding how mechanical forces interact with collective cell behavior at the tissue level. Modeling these complex phenomena often leads to nonlinear, strongly coupled partial differential equations (PDEs) that present significant computational challenges.

Traditional numerical methods like finite difference, finite volume, and finite element methods face difficulties when dealing with:

- Complex geometric shapes
- Mesh generation complexities
- Limited generalization capabilities
- Computational domain limitations

While deep learning approaches have emerged as alternatives for solving PDEs without mesh generation, they often lack reliable error control and convergence guarantees.

This paper introduces the Runge-Kutta Random Feature Method (RK-RFM), which combines the Random Feature Method (RFM) for spatial discretization with the explicit Runge-Kutta method for temporal discretization. This hybrid approach aims to solve the nonlinear and strongly coupled multiphase flow problems of cells effectively while maintaining high accuracy in both space and time.

2 Problem Formulation

2.1 Random Feature Method (RFM) for Static Problems

The RFM was initially designed to solve static linear boundary-value problems of the form:

$$L\phi(x) = f(x), \quad x \in \Omega, \quad (1)$$

$$B\phi(x) = g(x), \quad x \in \partial\Omega, \quad (2)$$

where L and B are linear differential and boundary operators, respectively.

The key components of RFM include:

1. Domain decomposition: Divide Ω into M_p non-overlapping subdomains Ω_n with centers x_n .
2. Linear transformation: Map each Ω_n into $[-1, 1]^d$ using $\tilde{x} = \frac{1}{r_n}(x - x_n)$.
3. Partition of Unity (PoU): Define PoU functions w_n with support on Ω_n .
4. Random feature functions: Construct functions u_{nj} using two-layer neural networks:

$$u_{nj}(x) = \sigma(W_{nj}^T \tilde{x} + b_{nj}), \quad (3)$$

where σ is a nonlinear activation function, and W_{nj} and b_{nj} are randomly chosen from uniform distribution $U(-R_m, R_m)$.

5. Approximate solution: Form a linear combination:

$$\tilde{\phi}(x) = \left(\sum_{n=1}^{M_p} w_n(x) \sum_{j=1}^{J_n} u_{nj}^1 u_{nj}^1(x), \dots, \sum_{n=1}^{M_p} w_n(x) \sum_{j=1}^{J_n} u_{nj}^{d_\phi} u_{nj}^{d_\phi}(x) \right)^T, \quad (4)$$

where $u_{nj}^i \in \mathbb{R}$ are unknown coefficients to be determined.

The RFM determines these coefficients by minimizing the loss function through linear least squares, using rescaling parameters to balance equation and boundary condition weights.

2.2 Multiphase Flow Problem of Cells

The paper focuses on a time-dependent multiphase flow problem that models cells as active deformable droplets:

$$\partial_t \phi_i = -v_i \cdot \nabla \phi_i - \frac{\delta F}{\delta \phi_i} = F(x, t, \phi(x, t)), \quad x, t \in \Omega \times (0, T], \quad i = 1, \dots, d_\phi, \quad (5)$$

$$B\phi(x, t) = g(x, t), \quad x, t \in \partial\Omega \times [0, T], \quad (6)$$

$$\phi(x, 0) = h(x), \quad x \in \Omega, \quad (7)$$

where ϕ_i represents the i -th cell, with $\phi_i = 1$ and $\phi_i = 0$ indicating the interior and exterior of the cell, respectively. The cell boundary is defined at $\phi_i = \frac{1}{2}$.

The free energy F is defined as $F = F_{CH} + F_{area} + F_{rep}$, where:

$$F_{CH} = \sum_i \int \frac{c}{k} (\phi_i^2 (1 - \phi_i)^2 + k^2 (\nabla \phi_i)^2) dx, \quad (8)$$

$$F_{area} = \sum_i \lambda \left(1 - \frac{1}{\pi R^2} \int \phi_i^2 dx \right)^2, \quad (9)$$

$$F_{rep} = \sum_i \sum_{j \neq i} \frac{j}{k} \int \phi_i^2 \phi_j^2 dx. \quad (10)$$

These terms represent:

- F_{CH} : Cahn-Hilliard free energy maintaining cell interface stability
- F_{area} : Soft regulation for cell area, targeting area close to πR^2
- F_{rep} : Penalty for cell overlap

The total velocity v_i of each cell is determined by the force balance equation:

$$\eta v_i = F_i^{int}, \quad (11)$$

where η is a substrate friction coefficient and F_i^{int} is the total force acting on the cell interface:

$$F_i^{int} = \int \phi_i \nabla \cdot \sigma^{tissue} dx = - \int \sigma^{tissue} \cdot \nabla \phi_i dx. \quad (12)$$

The tissue stress tensor σ^{tissue} is decomposed into passive and active stresses:

$$\sigma^{tissue} = -PI - \zeta Q, \quad (13)$$

where ζ represents activity strength, and P and Q are tissue pressure and nematic tensor.

3 Runge–Kutta Random Feature Method (RK-RFM)

3.1 Algorithm Description

The RK-RFM combines the RFM for spatial discretization with the explicit Runge-Kutta method for temporal discretization.

Given a time partition $[0, T] : 0 = t_0 < t_1 < \dots < t_K = T$, the explicit RK scheme is:

$$D_1(x) = F(x, t_k, \phi_{t_k}(x)), \quad (14)$$

$$D_2(x) = F(x, t_k + c_2 \Delta t, \phi_{t_k}(x) + a_{21} \Delta t D_1(x)), \quad (15)$$

$$\vdots \quad (16)$$

$$D_p(x) = F(x, t_k + c_p \Delta t, \phi_{t_k}(x) + \Delta t \sum_{i=1}^{p-1} a_{pi} D_i(x)), \quad (17)$$

$$\phi_{t_{k+1}}(x) = \phi_{t_k}(x) + \Delta t \sum_{i=1}^p b_i D_i(x) \equiv \phi_{t_k}(x) + \Delta t H(x, t_k, \phi_{t_k}(x), \Delta t), \quad (18)$$

where a_{pi} , b_i , and c_i are coefficients of the p -th order RK method.

The approximate solution $\tilde{\phi}_{t_{k+1}}(x)$ is represented using the RFM:

$$\tilde{\phi}_{t_{k+1}}(x) = \left(\sum_{n=1}^{M_p} w_n(x) \sum_{j=1}^{J_n} u_{nj,t_{k+1}}^1 u_{nj,t_{k+1}}^1(x), \dots, \sum_{n=1}^{M_p} w_n(x) \sum_{j=1}^{J_n} u_{nj,t_{k+1}}^{d_\phi} u_{nj,t_{k+1}}^{d_\phi}(x) \right)^T. \quad (19)$$

The RK-RFM minimizes the loss:

$$\text{Loss}\{u_{nj,t_{k+1}}^i\} = \sum_{n=1}^{M_p} \sum_{q=1}^Q \|k_{n,q,t_{k+1}}(\tilde{\phi}_{t_{k+1}}(x_q^n) - \phi_{t_{k+1}}(x_q^n))\|_2^2 \quad (20)$$

$$+ \sum_{n=1}^{M_p} \sum_{x_q^n \in \partial\Omega} \|k_{n,b,t_{k+1}}(B\tilde{\phi}_{t_{k+1}}(x_q^n) - g(x_q^n, t_{k+1}))\|_2^2, \quad (21)$$

to find the optimal coefficients $U_{t_{k+1}} = (u_{nj,t_{k+1}}^i)^T$ of $\tilde{\phi}_{t_{k+1}}(x)$.

This leads to a linear system:

$$A_{t_{k+1}} U_{t_{k+1}} = f_{t_{k+1}}, \quad (22)$$

which is solved using linear least squares.

3.2 Implementation Strategies

1. **Non-automatic differentiation:** For efficiency, the derivative functions are manually derived before computation:

$$\nabla_x^l u(x, W, b) = \nabla_x^l \sigma(W^T \tilde{x} + b), \quad (23)$$

and stored for direct retrieval during computation, avoiding costly automatic differentiation.

2. **Parallelization:** The same set of random feature functions is used for different components of the solution:

$$A_{t_{k+1}} U_{t_{k+1}} = A_{t_{k+1}} [U_{t_{k+1}}^1, \dots, U_{t_{k+1}}^{d_\phi}] = [f_{t_{k+1}}^1, \dots, f_{t_{k+1}}^{d_\phi}] = f_{t_{k+1}}, \quad (24)$$

allowing the linear least squares problem to be solved once for all components.

3.3 Complete RK-RFM Algorithm

4 Theoretical Error Estimates

The paper provides rigorous error estimates for the RK-RFM, establishing its convergence properties. The key results include:

Lemma 1. *Let σ be a bounded nonconstant piecewise continuous activation function, $d_\phi = 1$, $G_M(x)$ be of the form of the RFM approximation, and $M = M_p J_n$ denote the degree of freedom. Then, for any $f \in C(\Omega)$, there exists a sequence of $G_M(x)$ such that:*

$$\lim_{M \rightarrow \infty} \|G_M(x) - f(x)\|_{L^2} = 0, \quad \forall x \in \Omega. \quad (25)$$

Lemma 2. *For the RFM, the numerical solution $\hat{U} = (\hat{u}_{nj}^i)^T$ and the optimal solution $\tilde{U} = (\tilde{u}_{nj}^i)^T$ satisfy:*

$$E_{du}[\|\hat{U} - \tilde{U}\|_{L^2}] \leq \frac{d}{\sqrt{d_\phi M_p J_n}}, \quad (26)$$

where d is related to the distribution of random parameters.

Algorithm 1 The Runge–Kutta Random Feature Method

Require: M_p, J_n, Q, T, K , RK coefficients a_{pi}, b_i, c_i, R_m, c

Ensure: Optimal $U_{t_1}, \dots, U_{t_K}$

- 1: Divide Ω into M_p non-overlapping subdomains Ω_n
 - 2: Sample Q collocation points $\{x_q^n\}$ in each Ω_n
 - 3: Compute the derivative functions $\nabla_x^l u(x, W, b)$
 - 4: **for** $k = 0, 1, 2, \dots, K - 1$ **do**
 - 5: Construct J_n random feature functions $u_{nj, t_{k+1}}$ on Ω_n
 - 6: Compute $\phi_{t_{k+1}}(x)$ according to the RK scheme
 - 7: Assemble the linear system and solve by least squares to obtain $U_{t_{k+1}}$
 - 8: Compute $\tilde{\phi}_{t_{k+1}}(x)$ using the RFM representation
 - 9: $\phi_{t_{k+1}}(x) := \tilde{\phi}_{t_{k+1}}(x)$
 - 10: **end for**
 - 11: **return** Optimal $U_{t_1}, \dots, U_{t_K}$
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Theorem 1. Let $\phi_{t_k}^e$ be the exact solution and ϕ_{t_k} be the solution to the explicit p -th order RK method. Let $\tilde{\phi}_{t_k}$ and $\hat{\phi}_{t_k}$ be the optimal and numerical RFM solutions. Under certain Lipschitz conditions and approximation assumptions, there exist constants $C_1, C_2 > 0$ such that:

$$E_{du}[\|\hat{\phi}_{t_k}(x) - \phi_{t_k}^e(x)\|_{L^2}] \leq e^{L(t_k - t_0)} \left(\frac{\Delta t L C_1}{\sqrt{d_\phi M_p J_n}} d + \sqrt{d_\phi} e \right) + C_2 \Delta t^p, \quad (27)$$

where the error consists of two parts:

$$e_1(\Delta t) = e^{L(t_k - t_0)} \left(\frac{\Delta t L C_1}{\sqrt{d_\phi M_p J_n}} d + \sqrt{d_\phi} e \right), \quad e_2(\Delta t) = C_2 \Delta t^p. \quad (28)$$

Remark 1. The error consists of two components: $e_1(\Delta t)$ represents the error from RFM and optimization (inversely proportional to Δt), while $e_2(\Delta t)$ is the error from the RK method (proportional to Δt^p). Ideally, Δt should be chosen to balance these errors as $e_1(\Delta t) \approx e_2(\Delta t)$.

5 Numerical Experiments

5.1 Convergence Tests

The paper first validates the convergence of the RK-RFM using a test problem with known exact solution:

$$\phi_1(x, t) = \sin(x_1) \sin(x_2) e^{-t}, \quad (29)$$

$$\phi_2(x, t) = \cos(x_1) \cos(x_2) e^{-t}, \quad (30)$$

with appropriate initial and boundary conditions.

The method achieves second-order convergence in time when using the second-order RK scheme. Key findings include:

- For the **tanh** activation function, when $\Delta t = 5 \times 10^{-4}$, the error reaches minimum as $e_1(\Delta t) \approx e_2(\Delta t)$.
- For the **cos** activation function, minimum error occurs at $\Delta t = 5 \times 10^{-5}$.
- The error decreases as the number of random feature functions J_n and subdomains M_p increase, confirming the universal approximation capability.
- Increasing the number of collocation points Q also reduces error by decreasing the variance in random feature performance.

5.2 Multiphase Flow of Cells

The method was applied to simulate the multiphase flow problem of cells in a domain $\Omega = (0, 200)^2 \times (0, T]$ with initial conditions:

$$\phi_i(x, 0) = \frac{1 + \tanh(r - \|x - C_i\|_{L^2})}{2}, \quad i = 1, \dots, d_\phi, \quad (31)$$

and periodic boundary conditions. The simulation involved 240 cells.

The simulation parameters included:

- Physical parameters: $c = 0.01$ (elasticity), $\lambda = 3$ (area constraint), $j = 0.1$ (repulsion), $k = 2.5$ (interface width), $R = 8.0$ (target radius), $\eta = 2$ (friction), $\zeta = 0.005$ (activity)
- Algorithm parameters: $\Delta t = 0.1$, $M_p = 4 \times 4$, $Q = 40 \times 40$, $J_n = 600$

Key observations from the simulations:

1. Cells gradually increase in size due to the area constraint term F_{area} .
2. When cells approach each other, the repulsion term F_{rep} pushes them apart.
3. The Cahn-Hilliard term F_{CH} maintains stable cell interfaces.
4. Cells arrange into more ordered structures as the system evolves toward steady state.
5. Larger elasticity parameter c leads to rounder cells with greater resistance to deformation.
6. Larger repulsion parameter j increases spacing between cells.

Additional investigations examined the system's behavior under varying activity parameters ζ :

- The root mean square velocity v_{rms} and nematic order S_{rms} develop nonzero values with increasing activity.
- An activity threshold emerges, below which the system returns to equilibrium, and above which it maintains large flows and increasing order.
- With negative activity ($\zeta < 0$), the system remains stable as interface interactions restore the stable state.

6 Implications and Advantages

The RK-RFM offers several significant advantages for solving multiphase flow problems:

1. **High Accuracy:** Maintains high numerical accuracy in both space and time domains.
2. **Mesh-Free:** Eliminates the need for mesh generation, making it well-suited for complex geometric configurations.
3. **Nonlinear Problem Handling:** Effectively solves PDEs with strong nonlinearity that traditional methods struggle with.
4. **Computational Efficiency:**

- Non-automatic differentiation strategy accelerates computation
 - Parallelization through shared random feature functions saves computational resources
 - Achieves higher accuracy with fewer degrees of freedom compared to traditional methods
5. **Theoretical Foundation:** Provides rigorous error estimates and convergence analysis, addressing a key limitation of many machine learning-based PDE solvers.
 6. **Biological Applications:** Successfully captures complex cell dynamics, including:
 - Growth and area maintenance
 - Cell-cell repulsion
 - Interface stability
 - Collective migration and ordering
 - Activity-dependent phase transitions

The method's balance of accuracy, efficiency, and theoretical foundations makes it a promising approach for multiscale biological simulations and other complex PDE systems that exhibit strong nonlinearities and coupling effects.